A FINITE MORPHISM THAT IS AN ISOMORPHISM OFF FINITELY MANY POINTS FACTORS INTO GLUEINGS OF PAIRS OF POINTS AND SQUISHINGS OF TANGENT VECTORS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the skyscraper decomposition, the induction framework with intermediate subalgebras, the multi-point splitting, the Nakayama reduction at a single point, and the terminal k x k versus k[epsilon] classification are all retained verbatim.

Editorial note. The label lemma-factor-almost-isomorphism names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The two elementary modifications. Same chapter, Examples example-glue-points and example-squish-tangent-vector. Define the glueing of two points and the squishing of a tangent vector, the two step types of the factorization.
- Rings with exactly two subalgebras. Same chapter, Lemma lemma-no-in-between-over-k. Classifies a finite k-algebra with exactly two k-subalgebras as k x k or k[epsilon], the two terminal cases of the induction.

Source reference. Original reference 0C1J: sources/stacks-curves/pinned-original-source.tex, source lines 3202–3202. Prerequisite pointer invoked at proof line(s) 3267; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1K: sources/stacks-curves/pinned-original-source.tex, source lines 3228–3228. Prerequisite pointer invoked at proof line(s) 3268; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1I: sources/stacks-curves/pinned-original-source.tex, source lines 3183–3183. Prerequisite pointer invoked at proof line(s) 3317; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Lemma 0.1. *Let k be an algebraically closed field. Let $f : X' \to X$ be a finite morphism of algebraic k-schemes such that $\mathcal{O}_X \subset f_*\mathcal{O}_{X'}$ and such that f is an isomorphism away from a finite set of points. Then there is a factorization*

$$X' = X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

such that each $X_i \to X_{i-1}$ is either the glueing of two points or the squishing of a tangent vector (see Examples 0C1J and 0C1K).

Proof. Let $U \subset X$ be the maximal open set over which f is an isomorphism. Then $X \setminus U = \{x_1, \ldots, x_n\}$ with $x_i \in X(k)$. We will consider factorizations $X' \to Y \to X$ of f such that both morphisms are finite and

$$\mathcal{O}_X \subset g_*\mathcal{O}_Y \subset f_*\mathcal{O}_{X'}$$

where $g : Y \to X$ is the given morphism. By assumption $\mathcal{O}_{X,x} \to (f_*\mathcal{O}_{X'})_x$ is an isomorphism onless $x = x_i$ for some i. Hence the cokernel

$$f_*\mathcal{O}_{X'}/\mathcal{O}_X = \bigoplus \mathcal{Q}_i$$

is a direct sum of skyscraper sheaves $\mathcal{Q}_i$ supported at $x_1, \ldots, x_n$. Because the displayed quotient is a coherent $\mathcal{O}_X$-module, we conclude that $\mathcal{Q}_i$ has finite length over $\mathcal{O}_{X,x_i}$. Hence we can argue by induction on the sum of these lengths, i.e., the length of the whole cokernel.

If $n > 1$, then we can define an $\mathcal{O}_X$-subalgebra $\mathcal{A} \subset f_*\mathcal{O}_{X'}$ by taking the inverse image of $\mathcal{Q}_1$. This will give a nontrivial factorization and we win by induction.

Assume $n = 1$. We abbreviate $x = x_1$. Consider the finite k-algebra extension

$$A = \mathcal{O}_{X,x} \subset (f_*\mathcal{O}_{X'})_x = B$$

Note that $\mathcal{Q} = \mathcal{Q}_1$ is the skyscraper sheaf with value B/A. We have a k-subalgebra $A \subset A + \mathfrak{m}_A B \subset B$. If both inclusions are strict, then we obtain a nontrivial factorization and we win by induction as above. If $A + \mathfrak{m}_A B = B$, then $A = B$ by Nakayama, then f is an isomorphism and there is nothing to prove. We conclude that we may assume $B = A + \mathfrak{m}_A B$. Set $C = B/\mathfrak{m}_A B$. If C has more than 2 k-subalgebras, then we obtain a subalgebra between A and B by taking the inverse image in B. Thus we may assume C has exactly 2 k-subalgebras. Thus $C = k \times k$ or $C = k[\epsilon]$ by Lemma 0C1I. In this case f is correspondingly the glueing two points or the squishing of a tangent vector. □